Exercise: 31

Test for Lipids – Saponification test

Aim:

To detect the presence of fats (lipid) in different plants and animal materials.

Basic Principle:

1. Lipids are esters of fatty acid and alcohol
2. Lipids are not soluble in water and soluble in organic solvent like benzene, ether and chloroform.
3. Major groups of lipids are triglycerides, phospholipids, Steroids and Waxes.
4. Soapy appearance due break down of ester bonds by NaOH.

Requirements:

Test tubes, test tube stands, NaOH, oil/ghee/butter.

Procedure:

1. Take 1 ml of sample solution in a test tube.
2. Add 1 ml of 5% NaOH and mix it well.
3. Appearance of soapy solution.

Procedure	Observation	Inference
1 ml of sample solution + 1ml 5% NaOH solution and mix it well.	Appearance of Soapy solution	Lipid is present

VI - Plant Physiology Experiments.

Exercise: 32

Potato osmoscope experiment

Aim: To prove osmosis by Potato osmoscope.

Requirements: Peeled potato tuber, concentrated sugar solution, water, beaker,

Procedure:

1. Take a peeled potato tuber and make a cavity inside with the help of a knife.
2. Fill the cavity with concentrated sugar solution and mark the initial level.
3. Place this setup in a beaker of pure water.
4. After 10 minutes observe the sugar solution level and record your findings

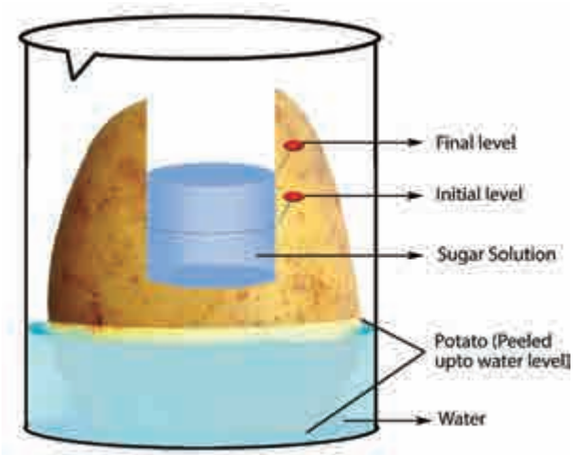


Figure 28 : Potato osmoscope experiment

Observation: The level of sugar solution increased in the cavity of the potato tuber.

Inference: It is proved that the increase in the level of sugar solution is due to osmosis.

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